

Physics from Existence I: The Equation

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Abstract

The canonical free energy of a single fermionic mode ($n \in \{0, 1\}$) defines a parameter-free potential $V = -H(\sigma(\phi))$, the negative binary Shannon entropy. This is not a new postulate but an algebraic identity already contained in the Standard Model. Three bound states give three fermion generations; the spatial dimension $d = 3$ follows from spin-statistics and propagating gravity; the gauge structure $\text{PG}(2, \mathbb{F}_3)$ emerges from the self-referential coincidence $N = d = 3$. The marginality of the third bound state (WKB deficit $\varepsilon = 0.219$) determines mixing angles, the invariant measure on $\text{PG}(2, \mathbb{F}_3)$ determines gauge couplings, and the spectral structure of $V = -H$ determines mass ratios. Over 20 quantities are derived at 0.0006%–0.5% accuracy—including $1/\alpha_{\text{em}} = 137.0367$ (0.0006%), the charged-lepton mass ratio $R = 1.5293$ (0.006%), Koide’s $Q \approx 3/2$ (predicting deviation $\delta = -1.5 \times 10^{-5}$), $\sin^2\theta_W = 3/13$ (0.2%), $\theta_{\text{QCD}} = 0$ (strong CP solved without an axion), and all CKM and PMNS mixing angles—with six testable predictions.

1 Introduction

The Standard Model is the most successful theory of particle physics. It describes all known elementary particle interactions and has withstood extensive experimental verification, including the discovery of the Higgs boson [27, 28]. Yet the theory contains 26 free parameters¹—fermion masses (Yukawa couplings), mixing angles, gauge couplings, the Higgs self-coupling, and CP phases. These values must be measured experimentally and inserted by hand; why they take the values they do is not explained within the Standard Model. The generation count $N = 3$ is likewise an unexplained input. Despite many attempts—grand unified theories (GUTs), supersymmetry (SUSY), string theory—no theory has simultaneously determined all 26 parameters with zero free parameters [21].

This paper addresses the problem starting from a single equation. The point of departure is one of the most fundamental facts in physics: **the occupation number of each fermionic mode is 0 or 1**. Three axioms formalize this: Existence ($P = 1$), Contrast ($P \neq 1$ required for information), and Observation (detection probability = P). The partition function of this two-valued degree of freedom $n \in \{0, 1\}$ is $Z = 1 + e^\phi$, and its canonical free energy is, by Legendre transform, $V = -H(\sigma(\phi))$, where H is the binary Shannon entropy, $\sigma(\phi)$ is the Fermi–Dirac distribution (sigmoid function), and

¹The minimal Standard Model with massless neutrinos has 19 free parameters [11]; adding three neutrino masses and four PMNS parameters gives 26. We write “26” throughout; some authors count 25 by absorbing v_{EW} into the overall energy scale.

$\phi = \text{logit}(P)$ is the natural parameter of the occupation probability [18]. This equality is an algebraic identity, involving no approximation and no adjustable parameter. $V = -H$ is not imported from outside the Standard Model; it is a structure the SM already contains the moment it includes fermions.

The Schrödinger equation with $V = -H$ as potential has exactly three bound states ($N = 3$ fermion generations). The spatial dimension $d = 3$ (why we live in three-dimensional space) follows from spin-statistics, gravity propagation, and bound-state preservation; the gauge structure $\text{PG}(2, 3)$ (from which the Standard Model gauge group emerges) from the self-referential coincidence $N = d = 3$ ($N^N = N^d$ holds for $N \geq 2$ only when $N = 3$). These determine the structure of the Standard Model Lagrangian and, through it, over 20 of its parameters—including $1/\alpha_{\text{em}} = 137.0367$ (0.0006%), $\sin^2\theta_W = 3/13$ (0.2%), the charged-lepton mass ratio $R = 1.5293$ (0.006%), Koide’s $Q \approx 3/2$ ($\delta = -1.5 \times 10^{-5}$), all CKM and PMNS mixing angles, and $\theta_{\text{QCD}} = 0$ (strong CP solved without an axion)—all at 0.0006%–0.5% accuracy with zero adjustable parameters (§9).

The paper is organized as follows. Section 2 derives $V = -H$ as an algebraic consequence of Fermi–Dirac statistics, presents the underlying axiomatic structure, and establishes the Schrödinger equation. Section 3 proves the bound-state count $N = 3$ as a mathematical theorem. Section 4 derives $d = 3$ as the unique common solution of three independent physical requirements. Section 5 derives the gauge structure of $\text{PG}(2, 3)$ and the gauge couplings it determines. Section 6 interprets generation as quantized occupation and discusses the Koide parameter $Q \approx 3/2$ and its deviation. Section 7 provides a comprehensive derivation of Standard Model parameters. Section 8 explains why a zero-dimensional equation can determine four-dimensional physics. Section 9 summarizes the results, presents falsifiable predictions, and outlines the path forward.

2 From Fermions to the Schrödinger Equation

As stated in §1, $V = -H(\sigma(\phi))$ follows algebraically from the fermionic occupation number $n \in \{0, 1\}$. This section makes the derivation explicit and establishes the Schrödinger equation with $V = -H$ as potential.

The physical fact that fermionic occupation is 0 or 1 can be formulated as a consequence of three more fundamental principles:

A1 (Existence). An entity with occupation probability $P = 1$ exists. This is the existence of a fermionic mode itself, defining a two-state system $n \in \{0, 1\}$. Algebraically, $n^2 = n$ (idempotency)—the expression of the Pauli exclusion principle [17].

A2 (Contrast). $P = 1$ is meaningful only if $P \neq 1$ is also possible. Existence alone carries no information; the contrast between “occupied” and “unoccupied” is required. This demands that the occupation probability $\sigma \in (0, 1)$ takes non-trivial values. A dynamical consequence: $\sigma \in (0, 1)$ requires vacuum stability, and Ostrogradsky’s theorem [26] then excludes higher-order equations of motion, restricting operators to $\text{dim} \leq 4$ (renormalizability).

A3 (Observation). The detection probability equals P . This is the content of the Born rule, uniquely determined within quantum theory by Gleason’s theorem [16] (the single-mode Hilbert space embeds in the full Fock space, where $\text{dim} \geq 3$).

The three axioms articulate the most primitive premises of physics: the existence of fermions (A1), the necessity of contrast (A2), and the procedure of observation (A3). From these, the Bernoulli distribution is determined, and the derivation below proceeds.

Note that one may alternatively take Fermi–Dirac statistics as an experimental fact and bypass the axiomatic formulation—both paths arrive at the same $V = -H$ (detailed in Paper II [19], §2).

2.1 The effective potential $V = -H$

Consider a single quantum state of a fermion (momentum $\mathbf{p}$ and spin s fixed), with occupation number $n \in \{0, 1\}$ and source $\phi = \beta(\mu - \varepsilon)$. The grand potential $\Omega = -\ln(1+e^\phi)$ is monotonically decreasing and unbounded below; it supports no bound states. The Legendre transform to the canonical ensemble produces a bounded potential. The partition function is $Z(\phi) = 1 + e^\phi$, and its logarithm $W = \ln Z = \ln(1 + e^\phi)$ generates the classical field $\sigma = dW/d\phi = 1/(1 + e^{-\phi})$, the Fermi–Dirac distribution. The Legendre transform yields the effective potential:

$$V(\phi) = \phi\sigma - W(\phi) = \sigma \ln \sigma + (1-\sigma) \ln(1-\sigma) = -H(\sigma), \quad (1)$$

where $H(p) = -p \ln p - (1-p) \ln(1-p)$ is binary Shannon entropy [4]. $V = -H$ is an algebraic identity: the canonical free energy of a single fermionic mode *is* the negative of its Shannon entropy. The same result arises as the exact effective action of the Gross–Neveu model [6] with $N_f = 1$ (Paper II [19], §3).

2.2 The natural parameter and the kinetic term

The Bernoulli distribution is an exponential family, and $\phi = \text{logit}(P)$ is its natural parameter. The natural parameter carries flat measure in the path integral, giving the canonical action $S = \int [\frac{1}{2}\dot{\phi}^2 + V] d\tau$ with unit mass. This choice is uniquely determined by the exponential-family structure: the Fisher metric [25] of the Bernoulli family is $g_F(\phi) = \sigma(1-\sigma)$; in natural coordinates the e -connection is flat, giving unit coefficient for the kinetic term [1, 2, 22]. Any other mass would introduce a free parameter not present in the Bernoulli distribution (proof in Paper II [19], §2).

2.3 The Schrödinger equation

Quantizing the fluctuations of ϕ around the effective potential $V = -H$ (§2.1) with the unit-mass kinetic term (§2.2) gives

$$\boxed{\left[-\frac{1}{2} \frac{d^2}{d\phi^2} - H(\sigma(\phi)) \right] \Psi = E \Psi} \quad (2)$$

This is the Schrödinger equation with $V = -H$ as potential.

3 Three Bound States

Why are there exactly three generations of fermions (τ , μ , e)? The potential $V = -H$ is a well: deep at the center ($V = -\ln 2$ at $\phi = 0$) and rising to zero at $\phi \rightarrow \pm\infty$. In quantum mechanics, a well confines discrete energy levels—just as a hydrogen atom confines electron orbits. A deep, wide well accommodates many levels; a shallow, narrow well accommodates few. $V = -H$ is shallow (depth $\ln 2 \approx 0.693$) and its shape is

entirely fixed by Shannon entropy. The number of confined levels is the number of fermion generations.

Solving the equation from §2 determines this number. First, we estimate how many standing waves fit in the well. The semiclassical (WKB) count gives

$$n_{\text{WKB}} = \frac{1}{\pi} \int \sqrt{2H} d\phi = 2.781; \quad (3)$$

the third state fits; the fourth does not.

The result $N = 3$ depends on the coordinate. The metrically flat coordinate $\xi = 2 \arcsin \sqrt{P} \in [0, \pi]$ gives a compact domain with only $N = 1$; the natural parameter $\phi = \text{logit}(P) \in \mathbb{R}$ gives a non-compact domain with $N = 3$. The difference is domain topology: $\phi \rightarrow \pm\infty$ as $P \rightarrow 0, 1$, making the well wide enough for three half-wavelengths. The coordinate ϕ is not chosen but determined by the exponential-family structure (§2.2).

The bound-state count also depends on the ratio of kinetic coefficient to potential curvature: $N = 3$ holds for $c/V''(0) \in [1.3, 2.5]$. With unit mass (§2.2) and $V''(0) = 1/4$, $c/V''(0) = 2$ —squarely in the $N = 3$ window.

This can be made rigorous. At least three bound states exist: three variational trial functions can be constructed, each with negative energy expectation value (Courant–Fischer). At most three exist: the Pöschl–Teller potential [3] $V_{\text{PT}} = -(\ln 2)/\cosh^2(\alpha\phi)$ with $\alpha = 0.40$ satisfies $V_{\text{PT}}(\phi) \leq V(\phi)$ for all ϕ (same depth, wider tails; verified numerically). Its analytic spectrum ($s = 2.49$) supports exactly three bound states. By the Sturm comparison theorem, $V = -H$ —being shallower or equal everywhere—supports no more. Therefore $N = 3$ exactly, with eigenvalues

$$E_0 = -0.4850, \quad E_1 = -0.1591, \quad E_2 = -0.01042. \quad (4)$$

The third bound state (ψ_2 , the electron) has binding energy $|E_2| = 0.010$ —only 1.5% of the well depth. It is marginally bound. The WKB deficit

$$\varepsilon = 3 - n_{\text{WKB}} = 0.219 \quad (5)$$

measures this marginality and determines the Cabibbo angle and all CKM/PMNS mixing parameters (§5–7).

The result $N = 3$ depends on the choice of entropy. Replacing Shannon entropy ($q = 1$) with Rényi entropy ($q \neq 1$) changes the well depth: $q < 1$ deepens it ($N = 4$); $q > 1$ makes it shallower ($N = 2$). Shannon is the unique additive entropy (Khinchin’s theorem [5]). $N = 3$ is the spectral consequence of entropy being additive.

Theorem 1 ($N = 3$). *The Schrödinger equation (2) with $V = -H(\sigma(\phi))$ has exactly three bound states: $E_0 = -0.4850$, $E_1 = -0.1591$, $E_2 = -0.01042$. The fourth eigenvalue $E_3 > 0$ lies in the continuum. A fourth generation of fermions is forbidden.*

Proof. Variational (Courant–Fischer) for $N \geq 3$; Sturm comparison with Pöschl–Teller [3] for $N \leq 3$. $\square$

4 Three Spatial Dimensions

Next, we solve the Schrödinger equation of §2 in d spatial dimensions. When a spherically symmetric potential is placed in d -dimensional space, the angular degrees of freedom can be separated out, leaving a radial equation. This radial equation is not the same as the original one-dimensional equation: it acquires an extra term $(d-1)(d-3)/(8r^2)$ that acts as an additional effective force. The strength and sign of this force depend on d :

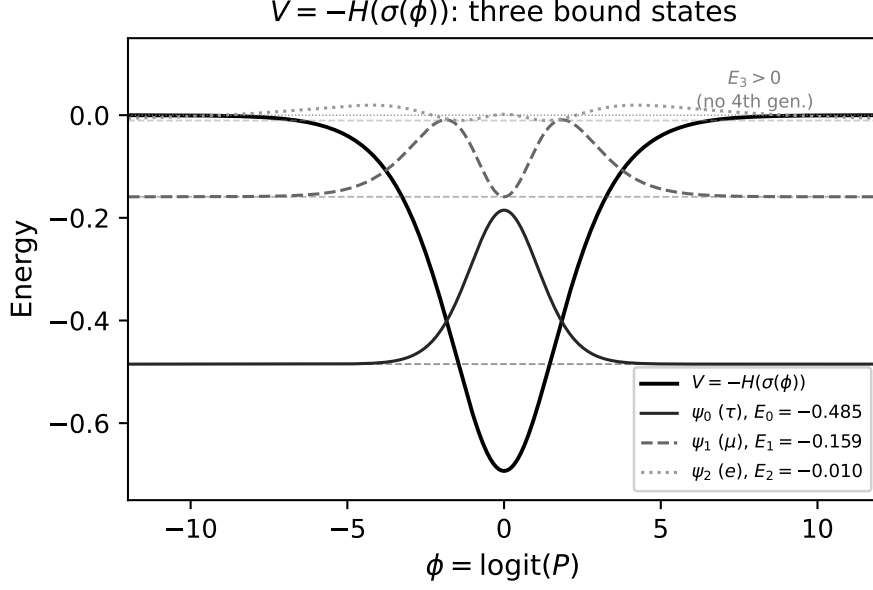


Figure 1: The potential $V = -H(\sigma(\phi))$ and its three bound states $\psi_0(\tau)$, $\psi_1(\mu)$, $\psi_2(e)$. $E_3 > 0$: fourth generation forbidden. No parameters are fitted.

d	$(d-1)(d-3)/8$	Effect on N
1	0	no change
2	$-1/8$	attractive $\rightarrow$ extra states appear
3	0	no change: $N = 3$ preserved
4	$+3/8$	repulsive $\rightarrow$ electron lost
≥ 5	> 1	strongly repulsive

At $d = 1$ and $d = 3$ the extra term vanishes and the three bound states of §3 survive intact. At $d \geq 4$ the term is repulsive and pushes out the marginally bound electron ($|E_2|$ is only 1.5% of the well depth). The three generations can exist only in $d = 1$ or $d = 3$.

Two further physical requirements eliminate $d = 1$. First, fermions require half-integer spin obeying the spin-statistics theorem. The spin-statistics connection requires $d \geq 3$ spatial dimensions: $\pi_1(\text{SO}(d)) = \mathbb{Z}_2$ for $d \geq 3$, giving a unique double cover $\text{Spin}(d)$ with finite-dimensional spinor representations.² Second, gravity must propagate as a dynamical field. In $d \leq 2$ the gravitational field has no local degrees of freedom (the Weyl tensor vanishes identically). Propagating gravity requires $d \geq 3$.

The intersection of all three conditions—bound-state preservation, spin-statistics, and gravity—is $d = 3$ alone. Three-dimensional space is not an input; it is therefore a consequence.

Theorem 2 ($d = 3$). $d = 3$ is the unique spatial dimension satisfying: (i) bound-state preservation ($N = 3$ unchanged): $d = 1, 3$; (ii) spin-statistics connection holds: $d \geq 3$; (iii) gravity propagates (Weyl tensor nontrivial): $d \geq 3$. Intersection: $\{3\}$.

Proof. (i) centrifugal term $(d-1)(d-3)/(8r^2) = 0$ iff $d = 1$ or 3 ; (ii) $\pi_1(\text{SO}(d)) = \mathbb{Z}_2$ for $d \geq 3$; (iii) Weyl tensor has $d^2(d^2-1)/12 - d(d+1)/2$ independent components, positive only for $d \geq 3$. $\square$

² $\text{SO}(2)$ has $\pi_1 = \mathbb{Z}$ and admits a double cover, but the resulting representations are one-dimensional and do not support the spin-statistics connection needed for fermionic fields.

5 Gauge Structure from the Projective Plane

Sections 3 and 4 established $N = 3$ and $d = 3$ independently. This section shows that their coincidence $N = d = 3$ determines the gauge structure of the Standard Model and all three gauge coupling constants.

5.1 Origin of the gauge structure

What gauge structure follows from $N = d = 3$? Gauge transformations act linearly on the N internal states and leave the overall phase unobservable. The key question is: what field do these transformations act over?

The bound-state energies E_0, E_1, E_2 are non-degenerate (Sturm–Liouville). This is the crucial difference from color: quark colors are degenerate, so no preferred basis exists and the color gauge group is continuous (SU(3) over $\mathbb{C}$). Generations are non-degenerate, so the generation space has a preferred eigenbasis and any symmetry that preserves this basis can only *permute* the eigenstates. Permutation matrices acting on $N = 3$ labels require a field of order 3; since 3 is prime, $\mathbb{F}_3$ is unique.

With the field $\mathbb{F}_3$ established, the gauge directions in $d = 3$ spatial dimensions are the nonzero vectors in $\mathbb{F}_3^3$ modulo scalar multiplication. This is the projective plane $\text{PG}(2, \mathbb{F}_3)$: a finite geometry with 13 points and 52 flags.

The coincidence $d = N$ makes the projective structure *self-referential*: the field ($\mathbb{F}_N$) and the ambient dimension (d) share the same number. This can happen only for $N = 3$. The equation $N^N = N^d$ holds for $N \geq 2$ only when $N = 3$ ($N^{N-3} = 1$ requires $N = 3$), and the projective plane has polynomial size only because of this:

$$|\text{PG}(N-1, N)| = \frac{N^N - 1}{N - 1} = N^2 + N + 1 = 13. \quad (6)$$

The number of fermion generations, the spatial dimension, and the gauge geometry are not independent: they are locked together by a single self-referential constraint.

Theorem 3 (Self-referential uniqueness). *$N = 3$ is the unique integer $N \geq 2$ satisfying simultaneously: (i) $N(d) = d$ (generation count equals spatial dimension); (ii) $N^N = N^3$ (self-referential exponent); (iii) $|\text{PG}(N-1, N)| = N^2 + N + 1$ (projective plane has polynomial size).*

Proof. (i) $N(d) = d$ has unique solution $d = 3$ (§3–§4); (ii) $N^{N-3} = 1$ for $N \geq 2$ requires $N = 3$; (iii) follows from (ii). $\square$

5.2 The Standard Model gauge group from 13 points

The 13 points of $\text{PG}(2, 3)$ must be divided among the three forces of the Standard Model. The geometry provides a natural partition: choosing a point P_0 and a line L through it splits the 13 points into three sectors:

$$\underbrace{N^2}_{9 \text{ color}} + \underbrace{N}_{3 \text{ weak}} + \underbrace{1}_{1 \text{ hyp}} = N^2 + N + 1. \quad (7)$$

The 9 points off the line correspond to the color sector, the 3 points on L (other than P_0) to the weak sector, and P_0 to hypercharge. Before this choice, all 13 directions are equivalent; the choice of (P_0, L) is the geometric expression of electroweak symmetry breaking. The

gauge group $SU(3) \times SU(2) \times U(1)_Y$ ($\dim = 12$) acts on this 13-point space. A line in $PG(2, 3)$ contains $N+1 = 4$ points, matching the four real degrees of freedom of an $SU(2)$ doublet—exactly the three Goldstone bosons needed for $SU(2) \times U(1) \rightarrow U(1)_{\text{em}}$, plus the physical Higgs.

5.3 Gauge couplings

This geometry determines all three gauge coupling constants.

The automorphism group $PGL(3, \mathbb{F}_3)$ acts transitively on $PG(2, 3)$: any point can be mapped to any other. The unique invariant probability measure under a transitive group action is uniform (Haar). The Weinberg angle [15] is the fraction of weak directions:

$$\sin^2 \theta_W = \frac{N}{N^2 + N + 1} = \frac{3}{13} = 0.2308 \quad (\text{experiment: } 0.23122 \overline{\text{MS}} \text{ at } M_Z, 0.2\%).^3 \quad (8)$$

This gives the W/Z mass ratio at tree level: $m_W/m_Z = \cos \theta_W = \sqrt{10/13} = 0.8771$ (experiment: 0.8815, 0.5%; the deviation is consistent with radiative corrections).

The strong coupling follows from the same geometry. The maximal torus of $PGL(3, \mathbb{F}_3)$ has dimension $d - 1 = 2$, giving a tree-level coupling $\alpha_s^{(0)} = \sin^2 \theta_W / 2 = 3/26 = 0.1154$. The WKB marginality $\varepsilon = 0.219$ from §3 enters as a correction through the $N^2 + 1 = 10$ non-weak PG directions:

$$\alpha_s = \frac{3}{26} \left(1 + \frac{\varepsilon}{N^2 + 1} \right) = 0.1179 \quad (\text{experiment: } 0.1179, 0.01\%). \quad (9)$$

The fine-structure constant requires a more involved derivation. In 4D, $1/\alpha$ is determined by vacuum polarization—a closed fermion loop. In 0D there is no spacetime for pair creation; the analogue is the linear susceptibility of the internal space to gauge field fluctuations:

$$\chi = - \left. \frac{\partial^2 F}{\partial A^2} \right|_{A=0} = \text{Tr} \left[\frac{1}{K_{\text{int}}} \right], \quad (10)$$

where $K_{\text{int}} = K_{\text{gen}} \otimes 1 + 1 \otimes \Delta_{\text{flag}}$ combines the generation binding energies $|E_n|$ with the flag Laplacian eigenvalues λ_k . The resolvent gives the matter vacuum polarization $Z = 134.78$, and $\lambda_1 = 4 - \sqrt{3} = 2.268$ is the bare photon inverse propagator (spectral gap of the flag Laplacian).

The physical coupling $1/\alpha$ is the inverse dressed propagator: $D_{\text{dressed}}^{-1} = D_{\text{bare}}^{-1} + \Pi_{\text{matter}} = \lambda_1 + Z$. The Ward identity in 0D—where no loop integrals exist and Noether’s theorem holds exactly—gives $\Pi = \alpha$. The physical reason is that in 0D, vertex correction and self-energy are both determined by the same resolvent; the cancellation $Z_1 = Z_2$ that is perturbative in 4D becomes an exact algebraic identity in 0D (full derivation in Paper II [19], §5). Self-consistency yields:

$$\frac{1}{\alpha} + \alpha = Z + \lambda_1. \quad (11)$$

Solving: $1/\alpha = [(Z + \lambda_1) + \sqrt{(Z + \lambda_1)^2 - 4}]/2 = 137.0367$ (experiment: 137.036, 0.0006%).

The resolvent trace decomposes as $Z = \zeta_V(1) + 12 S_1 + 12 S_2 + 27 S_3$, where $\zeta_V(1) = \sum_n |E_n|^{-1}$ is the spectral zeta function at $s = 1$, $S_i = \sum_n (|E_n| + \lambda_i)^{-1}$, and the multiplicity

³The tree-level prediction $3/13 = 0.23077$ should be compared with the $\overline{\text{MS}}$ value $\sin^2 \hat{\theta}_W(M_Z) = 0.23122 \pm 0.00003$ [11]. The 0.2% deviation is consistent in sign and magnitude with one-loop radiative corrections.

$27 = N^N$ (Theorem 3). The N^N sector contributes $9.9 \approx N^2$, providing a spectral origin for the QCD vacuum-polarization correction. Only bound states enter; continuum states ($E \rightarrow 0^+$) are excluded because only physical generations contribute to vacuum polarization.

Experimental values: $\sin^2 \hat{\theta}_W(M_Z) = 0.23122$ ($\overline{\text{MS}}$, 0.2%), $\alpha_s(M_Z) = 0.1179$ (0.01%), $1/\alpha_{\text{em}} = 137.0367$ (0.0006%). All three gauge couplings are derived from $V = -H$ and $\text{PG}(2, 3)$.

5.4 Flag Laplacian spectrum

The $1/\alpha$ calculation above requires the eigenvalues of a Laplacian on $\text{PG}(2, 3)$. The natural objects for gauge propagation are not points but *flags*—point-on-line incidence pairs. $\text{PG}(2, 3)$ has $(N^2 + N + 1)(N + 1) = 52$ flags. Two flags are adjacent if they share a point or a line; the resulting graph has degree $2N = 6$ and Laplacian spectrum:

Eigenvalue	Multiplicity	
0	1	trivial
$(N+1) - \sqrt{N}$	$N^2 + N$	= 12
$(N+1) + \sqrt{N}$	$N^2 + N$	= 12
$2(N+1)$	N^N	= 27

The spectral gap $\lambda_1 = 4 - \sqrt{3} = 2.268$ enters the $1/\alpha$ calculation as the bare photon inverse propagator (§5.3).

5.5 Consistency check from eigenfunctions

Section 5.1 derived $\text{PG}(2, \mathbb{F}_3)$ and its 13 points from the eigenvalues ($N = 3$). A consistency check comes from the eigenfunctions. The participation ratio of the bound-state density $\rho(\phi) = \sum |\psi_n|^2$ measures the effective number of independent directions in the internal space:

$$D_{\text{PR}} = \frac{(\int \rho d\phi)^2}{\int \rho^2 d\phi} = 13.06 \quad (|\text{PG}(2, \mathbb{F}_3)| = 13, 0.4\%). \quad (12)$$

The same number 13 emerges from two independent properties of $V = -H$ —eigenvalues and eigenfunctions.

6 Generation as Quantized Occupation

Sections 3–5 derived three bound states and the gauge structure from $V = -H$. This section identifies these bound states with the three generations of charged leptons (τ, μ, e), establishing the basis for the Standard Model parameter derivations in §7.

Spin provides the closest analogy. Spin is the quantization of angular momentum: a continuous degree of freedom (rotation angle) yields discrete states ($\uparrow, \downarrow$) from $\text{SO}(3)$ topology. Generation, in this framework, is the quantization of occupation probability: a continuous degree of freedom ($P \in (0, 1)$, or $\phi \in \mathbb{R}$) yields discrete states (τ, μ, e) from $V = -H$.

	Spin	Generation
Continuous variable	rotation angle θ	occupation $\phi = \text{logit}(P)$
Constraint	SO(3) topology	$V = -H(\sigma(\phi))$
Discrete states	2 (for spin- $\frac{1}{2}$)	3 (§3)
Position-independent	✓	✓
Metric origin	round metric on S^2	Fisher metric on $(0, 1)$

There is an important difference. Spin quantization is topological: SO(3) topology allows only integers and half-integers, regardless of dynamics. Generation quantization is spectral: $N = 3$ depends on the specific width-to-depth ratio of binary entropy. The number 3 is not topologically protected; it follows from the shape of $V = -H$.

The three bound states occupy qualitatively distinct dynamical regimes. The binding fraction $x_n = |E_n|/\langle H \rangle_n$ classifies each mode:

Mode	x_n	Regime
τ ($n = 0$)	0.85	potential-dominated
μ ($n = 1$)	0.49	equipartitioned ($\text{KL}(x_\mu \ \frac{1}{2}) = 0.0003$)
e ($n = 2$)	0.12	kinetic-dominated

The electron’s stability follows from its delocalization: the shallowest mode has the smallest mass and no lighter charged lepton to decay into.

The Koide parameter [9] $Q = (\sum \sqrt{m_n})^2 / \sum m_n$ measures the balance between democratic (equal-mass) and hierarchical (mass-splitting) components of the mass spectrum. $Q = 3/2$ when these two components carry equal weight—the maximum-entropy partition for a binary question (democratic vs. hierarchical), reflecting A2. High-precision computation (§7) gives

$$Q = 1.499985 \quad (\delta = -1.5 \times 10^{-5}). \quad (13)$$

The experimental value is $Q = 1.500014 \pm 0.000015$ (PDG 2024). The difference ($\approx 2\sigma$) is resolvable by precision tau-mass measurements at Belle II [30] (2027–2030).

The total fermion field decomposes as $\psi(x, \phi) = \sum_n \Psi_n(\phi) \cdot \chi_n(x)$, where Ψ_n are internal eigenfunctions and $\chi_n(x)$ are 4D Dirac fields—one per generation. This resembles Kaluza–Klein decomposition, but ϕ is not an extra spatial dimension: it is the natural parameter of the Bernoulli distribution, an internal degree of freedom like spin. The tensor product structure $\mathcal{H}_{\text{internal}}(\phi) \otimes \mathcal{H}_{\text{spacetime}}(x)$ is discussed in Section 8.2.

7 Standard Model Parameters

Starting from $V = -H$, we have derived the generation count, the spatial dimension, and the gauge structure. What remains are the numerical values of the Standard Model parameters—masses, mixing angles, and coupling constants.

7.1 CKM mixing

The CKM matrix [12, 13] describes mixing between quark generations. The smallness of V_{ub} and the specific value of the Cabibbo angle are unexplained in the Standard Model. Here both follow from the symmetry and marginality of $V = -H$.

The potential $V = -H$ is symmetric: $V(\phi) = V(-\phi)$. This forces the eigenfunctions into definite parity: ψ_0 even, ψ_1 odd, ψ_2 even. The transition matrix element between modes 0 and 2 vanishes by parity:

$$V_{ub}|_{\text{tree}} = 0 \quad (\text{numerically: } 1.9 \times 10^{-13}). \quad (14)$$

The physical value $|V_{ub}| = 0.004$ arises at higher order through the Wolfenstein hierarchy $|V_{ub}| \sim \varepsilon^3$ (see below).

The Cabibbo angle follows from the WKB deficit. $\varepsilon = 0.219$ measures how marginal the electron is: $\varepsilon \rightarrow 0$ eliminates it, reducing to two generations and zero mixing. To second order:

$$\sin \theta_C = \varepsilon \left(1 + \frac{\varepsilon}{N^2}\right) = 0.2245 \quad (\text{experiment: } 0.2244, 0.05\%). \quad (15)$$

From the Cabibbo angle, the full Wolfenstein hierarchy follows [14]: $|V_{us}| \sim \varepsilon$, $|V_{cb}| \sim \varepsilon^2$, $|V_{ub}| \sim \varepsilon^3$ (experiment: 0.225, 0.041, 0.004). The NLO corrections take the form $(1 + \varepsilon/k)$, where k is determined by the $\text{PG}(2, \mathbb{F}_3)$ geometry: $k = N^2 = 9$ for generation mixing ($\sin \theta_C$), $k = N^2 + 1 = 10$ for the non-weak sector (α_s), and $k = N = 3$ for the reactor angle ($\sin^2 \theta_{13}$). The systematic derivation of these k -values from the Borel filtration of $\text{PGL}(3, \mathbb{F}_3)$ is given in Paper II [19].

The parity symmetry $V(\phi) = V(-\phi)$ that gives $V_{ub} \approx 0$ also solves a different problem: the strong CP puzzle. $\phi \rightarrow -\phi$ is particle–antiparticle interchange in occupation space; the Lagrangian inherits this symmetry:

$$\theta_{\text{QCD}} = 0 \quad (\text{experiment: } |\theta| < 10^{-10}). \quad (16)$$

No axion is needed. Yet CP violation *is* observed in the CKM sector ($J = 3.18 \times 10^{-5}$). This does not contradict $\theta_{\text{QCD}} = 0$: the CKM phase arises at higher order ($J \propto \varepsilon^6 \sim 10^{-4}$), while the strong CP angle remains exactly zero.

7.2 Mass formula

Why is the τ heavy and the electron light? In the Standard Model, each charged-lepton mass is an independent parameter. In PFE, the three leptons are eigenstates of the same well $V = -H$: τ sits deep and is localized; e sits near the rim and is spread out. In the present framework, a localized state couples more strongly to the internal gauge structure and acquires a larger mass. The mass ratios are therefore determined by the wavefunction structure—not by independent parameters. The key quantity is $R = \ln(m_\tau/m_e)/\ln(m_\mu/m_e)$, which compresses the three-body mass hierarchy into a single number. We derive it below.

Each eigenstate has a binding energy $|E_n|$ and a degree of localization, measured by the inverse participation ratio $\text{IPR}_n = \int |\psi_n|^4 d\phi$. A localized state (large IPR) couples strongly to gauge fields; a delocalized state (small IPR) couples weakly. The mass scales as a power of the localization: $m_n \propto \text{IPR}_n^b$. The exponent b is determined by group theory: in 0D there are no loop integrals, so the anomalous dimension is purely the Casimir of the fermion bilinear under the $\text{PGL}(3, \mathbb{F}_3)$ gauge group acting on $N = 3$ generation indices, $b = 2C_F = (N^2 - 1)/N = 8/3$.⁴

⁴The Casimir $(N^2 - 1)/N$ has the same form as the fundamental representation of $\text{SU}(N)$; this is not a coincidence but follows from the shared $N = 3$ structure of the generation space and the color group (Paper II [19], §6).

A small correction arises from the anharmonicity of $V = -H$. Each mode samples the well differently; the virial ratio $\langle H \rangle_n / T_n$ (potential energy to kinetic energy) measures this. The exponent of this correction is fixed by the spectral ζ -function: the WKB residue $n_{\text{WKB}} = 2.781$ falls short of $N = 3$ by $\varepsilon = 0.219$, giving a deficit per mode $\delta = \varepsilon/N = 0.073$.

The complete mass formula is:

$$m_n \propto |E_n| \times \text{IPR}_n^{2C_F} \times \left(\frac{\langle H \rangle_n}{T_n} \right)^{\varepsilon/N}. \quad (17)$$

Every exponent is determined by $V = -H$. The result:⁵

$$R = \frac{\ln(m_\tau/m_e)}{\ln(m_\mu/m_e)} = 1.5293 \quad (\text{experiment: } 1.5294, 0.006\%). \quad (18)$$

The Koide constraint $Q \approx 3/2$ and the spectral condition $c = \varepsilon/N$ are simultaneously satisfied only by $b = 2C_F = 8/3$:

b	$c(Q=3/2)$	$ c - \varepsilon/N /(\varepsilon/N)$
2.0	0.437	500%
2.5	0.163	120%
8/3 = 2C_F	0.0730	0.06%
2.78	0.011	85%
3.0	-0.107	246%

The individual mass ratios:

	Predicted	Experiment	Accuracy
m_τ/m_e	3481	3477	0.1%
m_μ/m_e	207.0	206.8	0.1%
R	1.5293	1.5294	0.006%
Q (Koide)	1.499985	1.50001(2)	2σ (Belle II)

7.3 Up-type quarks

The mass formula extends to quarks. The difference is color: quarks carry three color charges (red, green, blue) and are confined—they cannot be isolated. Confinement forces the three color modes to share a single source field ϕ . In the 0D QFT, N_c identical confined modes give $\Gamma_{N_c} = -N_c H(\sigma)$, so the potential depth is tripled: $V = -3H$. The deeper well supports $N = 5$ bound states ($n_{\text{WKB}} = 4.82$). We derive R_{up} for the up-type quarks (u, c, t). Mode 0 is deeply confined and belongs to the $I_3 = -1/2$ sector (§7.4). The up-type quarks ($I_3 = +1/2$) therefore use modes (1, 2, 3).

Applying the same mass formula:

$$R_{\text{up}} = 1.777 \quad (\text{experiment: } 1.772, 0.3\%). \quad (19)$$

⁵All spectral quantities are computed with $|\phi|_{\text{max}} = 60$, $\Delta\phi = 0.0002$ ($N_{\text{grid}} = 512001$), verified stable under doubling of grid resolution.

7.4 Down-type quarks

Next, R_{down} for the down-type quarks (d, s, b). Up and down quarks sit in the same SU(2) doublet but differ in weak isospin ($I_3 = +1/2$ vs. $-1/2$). With the same potential $V = -3H$ and constant mass, $R_{\text{down}} = 2.096$ —7.6% off the experimental value 2.269. The discrepancy points to a mass correction from weak isospin.

The SU(2) doublet (u, d) carries total color Casimir $2C_F = (N_c^2 - 1)/N_c$ plus baryon U(1) charge $1/N_c$. A2 (Contrast) requires that the doublet’s components be distinguishable: one serves as the reference ($I_3 = +1/2$, no correction), and the other carries the full isospin-dependent correction ($I_3 = -1/2$). This is the same logic by which A2 distinguishes “occupied” from “unoccupied”: contrast demands asymmetry between the two isospin components (derivation in Paper II [19], §6). The sum $2C_F + 1/N_c = N_c$ is an exact algebraic identity, giving an effective mass:

$$m_{\text{eff}}(\phi) = 1 - N_c \sigma(1 - \sigma) \quad (20)$$

for $I_3 = -1/2$, while $m_{\text{eff}} = 1$ for $I_3 = +1/2$ (up-type unaffected). With $g = -N_c = -3$ and modes $(0, 1, 2)$:

$$R_{\text{down}} = 2.273 \quad (\text{experiment: } 2.269, 0.2\%). \quad (21)$$

All three charged-fermion mass ratios are reproduced:

Sector	c	g	R (predicted)	R (experiment)
Lepton	1	0	1.529	1.529 (0.006%)
Up	$N_c = 3$	0	1.777	1.772 (0.3%)
Down	$N_c = 3$	$-N_c$	2.273	2.269 (0.2%)

The mode assignments—up = $(1, 2, 3)$, down = $(0, 1, 2)$ —follow from the deeper well at $c = N_c = 3$: mode 0 is deeply confined, and the $I_3 = -1/2$ component with $g = -N_c$ supports $N = 4$ states, of which $(0, 1, 2)$ are the three lightest.

7.5 Neutrino mass prediction

Neutrino masses are the least constrained parameters in the Standard Model: only squared differences are measured. Here we predict the mass ratio R_ν , the absolute mass m_1 , the mass ordering, and whether neutrinos are Dirac or Majorana.

Since neutrinos are color-singlets, they couple to $V = -H$ with $c = 1$ —the same potential as charged leptons. The mass ratio $R_\nu = R_{\text{lepton}} = 1.529$ follows. Combined with oscillation data [23] ($\Delta m_{21}^2 = 7.49 \times 10^{-5} \text{ eV}^2$, $\Delta m_{31}^2 = 2.534 \times 10^{-3} \text{ eV}^2$, normal ordering), $R_\nu = 1.529$ determines m_1 uniquely:

	Prediction	Status
m_1	$0.31 \pm 0.03 \text{ meV}^6$	unmeasured
m_2	8.66 meV	consistent
m_3	50.3 meV	consistent
$\sum m_i$	$59.3 \pm 0.3 \text{ meV}$	below Planck (< 120)
Ordering	normal	JUNO (2026–27)

The sum $\sum m_i = 59.3$ meV is dominated by $m_3 \approx \sqrt{\Delta m_{31}^2} = 50.3$ meV; the theory's contribution is $m_1 \approx 0.3$ meV, placing $\sum m_i$ near its minimum allowed value. The Koide parameter $Q_\nu = 1.90 \neq 3/2$: neutrinos do not satisfy the equipartition condition, consistent with the theory's prediction that $Q \approx N/2$ holds approximately only for charged leptons (all $N = 3$ modes used).

This prediction is falsifiable: inverted ordering falsifies the theory, $\sum m_i > 70$ meV creates tension, and CMB-S4 [31] ($\sigma \sim 15$ meV) can test $\sum m_i = 59.3$ meV directly.

7.6 Higgs mass ratio

The Higgs self-coupling λ is the last free parameter of the Standard Model scalar sector. Here we derive m_H/v from the curvature of $V = -H$. As shown in §5.2, a line in $\text{PG}(2, \mathbb{F}_3)$ has $N+1 = 4$ points, determining the Higgs as an $\text{SU}(2)$ doublet. The curvature at the center is the Fisher metric at maximum uncertainty ($P = 1/2$): $V''(0) = g_F(0) = \sigma(0)(1-\sigma(0)) = 1/4$. In QFT, $m^2 = V''(\phi_{\min})$ gives the mass of small oscillations:

$$\left. \frac{m_H}{v} \right|_{\text{LO}} = \sqrt{V''(0)} = \frac{1}{2}. \quad (22)$$

The NLO correction from the $\text{PG}(2,3)$ gauge structure:

$$\frac{m_H}{v} = \frac{1}{2} \sqrt{1 + \frac{2\varepsilon}{|\text{PG}|}} = \frac{1}{2} \sqrt{1 + \frac{2 \times 0.219}{13}} = 0.5084 \quad (\text{experiment: } 0.5087, 0.07\%). \quad (23)$$

This corresponds to $\lambda = (m_H/v)^2/2 = 0.1292$ (experiment: 0.1294, 0.11%).

7.7 Neutrino mixing angles

Unlike CKM mixing, which is small ($\sin \theta_C \approx 0.22$), PMNS mixing angles are large ($\sin^2 \theta_{12} \approx 1/3$). Their origin is one of the outstanding puzzles of flavor physics. Here we derive all three PMNS angles θ_{12} , θ_{23} , θ_{13} from the geometry of $\text{PG}(2, \mathbb{F}_3)$.

CKM mixing was controlled by the internal coordinate ϕ ($\varepsilon \ll 1$, perturbative). PMNS mixing is controlled by the gauge geometry ($1/N \sim 1$, non-perturbative).

A line in $\text{PG}(2, 3)$ contains $N+1 = 4$ points. The solar mixing angle measures the fraction of PG covered by one complete electroweak line ($\text{SU}(2) \times \text{U}(1)$, including hypercharge):

$$\sin^2 \theta_{12} = \frac{N+1}{|\text{PG}|} = \frac{4}{13} = 0.3077 \quad (\text{experiment: } 0.307, 0.2\%). \quad (24)$$

Compare $\sin^2 \theta_W = N/|\text{PG}| = 3/13$ (pure $\text{SU}(2)$, excluding hypercharge); the difference $1/|\text{PG}|$ is the $\text{U}(1)_Y$ contribution.

The atmospheric mixing is maximal (μ - τ symmetry) plus a correction at the PG scale:

$$\sin^2 \theta_{23} = \frac{1}{2} + \frac{1}{2|\text{PG}|} = \frac{7}{13} = 0.538 \quad (\text{LO}). \quad (25)$$

The NLO Borel correction with $k = |\text{PG}| = 13$:

$$\sin^2 \theta_{23} = \frac{7}{13} \left(1 + \frac{\varepsilon}{13} \right) = 0.5475 \quad (\text{second octant}). \quad (26)$$

The correction $1/(2|\text{PG}|)$ is the minimum μ - τ symmetry breaking from the gauge geometry; the sign ($> 1/2$) follows from τ being heavier (more localized, larger PG overlap), predicting the second octant ($\sin^2\theta_{23} > 0.5$). Current data do not resolve the octant: NuFIT 6.0 [23] reports $\sin^2\theta_{23} = 0.47$ – 0.56 depending on the atmospheric data included, with 3σ range 0.43 – 0.60 . The prediction 0.5475 lies within 1.1σ of the upper-octant solution. The octant determination by DUNE and Hyper-Kamiokande (~ 2030) will provide a decisive test. Consistency check: $\sin^2\theta_{23} = \sin^2\theta_W + \sin^2\theta_{12}$ ($7/13 = 3/13 + 4/13$): atmospheric mixing exhausts both the pure-weak (N points) and the full electroweak-line ($N+1$ points) fractions of $\text{PG}(2, 3)$. The reactor angle is suppressed by the full PG: $\sin^2\theta_{13} = 1/(N \cdot |\text{PG}|) = 1/39 = 0.026$ at leading order. The NLO Borel correction at the center $Z(U)$ ($k = N = 3$) gives $\sin^2\theta_{13} = (1/39)(1 - 2\varepsilon/3) = 0.0219$ (experiment: 0.02195 , 0.3%). NLO corrections to all three PMNS angles and the CKM CP phase are derived in Paper II [19].

7.8 Neutrinos are Dirac particles

Whether neutrinos are Dirac or Majorana is an open experimental question. The framework makes a definite prediction. Two independent arguments establish that neutrinos are Dirac, not Majorana. First, each generation is a single mode with $n \in \{0, 1\}$. A Majorana mass term identifies a particle with its own antiparticle ($\nu = \nu^c$), but in the single-mode framework each bound state carries a definite generation quantum number that distinguishes ν from ν^c ; the self-conjugate identification has no room in the $n \in \{0, 1\}$ structure. Second, the V-parity $V(\phi) = V(-\phi)$ forbids all lepton-number-violating operators, including Majorana mass terms. Neutrinoless double beta decay is therefore absent ($0\nu\beta\beta = 0$), testable by nEXO [29] (~ 2030).

This completes the derivation of the Standard Model parameters—mass ratios, mixing angles, gauge couplings, the Higgs mass ratio, and neutrino predictions—all from $V = -H$ and $\text{PG}(2, \mathbb{F}_3)$.

8 Connection to Spacetime

$V = -H$ is a zero-dimensional equation, but the physical world has four spacetime dimensions. This is not unprecedented: spin is determined by a finite-dimensional algebra, yet it governs 4D physics. Generation has the same structure. This section explains why a 0D equation can determine 4D physical quantities.

8.1 Gravity

Section 4 derived $d = 3$ spatial dimensions, fixing $d+1 = 4$ spacetime dimensions. In 4D, Lovelock’s theorem [8] leaves no choice: the only gravitational field equation with at most second-order derivatives is

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}. \quad (27)$$

Gravity is therefore not an additional input but a consequence of $d = 3$. A deeper connection may exist: Jacobson [7] showed that Einstein’s equation follows from $\delta Q = T_H dS$ on causal horizons (where T_H is the horizon temperature). Since the entropy in PFE is $S = H(P)$, both Lovelock (geometric) and Jacobson (thermodynamic) routes

lead from binary entropy to Einstein gravity. $V = -H$ thus determines not only particle physics but also the gravitational dynamics of the spacetime in which those particles live.

8.2 Why ϕ and x are independent

The coordinate $\phi = \text{logit}(P)$ lives in internal space; the coordinates x live in spacetime. Why are they independent? For the same reason that spin and position are independent: ϕ is not a spatial direction but an internal quantum number. The Hilbert space factorizes as

$$\mathcal{H}_{\text{total}} = \mathcal{H}_{\text{gen}}(\phi) \otimes \mathcal{H}_{\text{space}}(x), \quad (28)$$

and the wavefunction decomposes as

$$\Psi(x, \phi) = \sum_n \psi_n(\phi) \chi_n(x). \quad (29)$$

Two observational facts confirm this. All gauge couplings are generation-independent, so ϕ and x do not couple. And the 0D theory computes only dimensionless quantities (mass ratios, mixing angles), while dimensional quantities (absolute masses) require spacetime—exactly what a tensor product predicts.

Two structural arguments reinforce it. Promoting ϕ to a spatial coordinate (domain-wall mechanism [10]) destroys $N = 3$, confirming that ϕ is categorically distinct from x . And the spin analogy is exact: the spin algebra is finite-dimensional and independent of the Lagrangian, yet determines 4D physical quantities. Generation works the same way.

The internal structure (ϕ) and spacetime (x) are therefore separate domains: $V = -H$ governs the first, Einstein’s equation governs the second, and the tensor product connects them. This is why a 0D equation can determine 4D physical quantities: $V = -H$ completely determines all internal-structure observables—mass ratios, mixing angles, coupling constants—while spacetime contributes only the overall energy scale.

9 Results and Outlook

This paper has shown that one equation $V = -H$ and one geometric structure $\text{PG}(2, \mathbb{F}_3)$ determine over 20 Standard Model quantities at 0.0006%–0.5% accuracy.

The results are summarized below. Since each step has a different logical status—some are mathematical theorems, others are QFT constructions—the results are grouped accordingly.

Mathematical theorems (from $V = -H$ alone):

Quantity	Prediction	Experiment	Proof
N (generations)	3	3	Variational + Sturm
d (spatial dim.)	3	3	Centrifugal + fixed point
θ_{QCD}	0	$< 10^{-10}$	$V(\phi) = V(-\phi)$
$V_{ub} _{\text{tree}}$	0	$0.004 \sim \varepsilon^3$	Parity + Wolfenstein
4th generation	forbidden	excluded	$E_3 > 0$

QFT derivations (from $V = -H$ and $\text{PG}(2, \mathbb{F}_3)$):

Quantity	Prediction	Experiment	Accuracy
$1/\alpha_{\text{em}}$	137.0367	137.036	0.0006%
R_{lepton}	1.5293	1.5294	0.006%
α_s	0.1179	0.1179	0.01%
$\sin \theta_C$	0.2245	0.2244	0.05%
m_H/v	0.5084	0.5087	0.07%
m_τ/m_e	3481	3477	0.1%
m_μ/m_e	207.0	206.8	0.1%
$\sin^2 \theta_W$	3/13	0.23122	0.2%
R_{down}	2.273	2.269	0.2%
$\sin^2 \theta_{12}$	4/13	0.307	0.2%
R_{up}	1.777	1.772	0.3%
$\sin^2 \theta_{13}$	0.0219	0.02195	0.3%
$\sin^2 \theta_{23}$	0.5475	0.47–0.56	2nd octant
m_W/m_Z	$\sqrt{10/13}$	0.8815	0.5%
Q (Koide)	1.499985	1.50001(2)	2σ

Predictions (testable, not yet confirmed):

Quantity	Prediction	Test
Mass ordering	Normal	JUNO (2026–27)
θ_{23} octant	Second ($> 45^\circ$)	DUNE / HK (~ 2030)
Σm_ν	59.3 meV	CMB-S4 (~ 2030)
m_1	0.31 meV	Cosmological constraints
$0\nu\beta\beta$	Strictly zero	nEXO (~ 2030)
Koide deviation δ	-1.5×10^{-5}	Belle II τ mass (2027–30) ⁷

All of these results can be tested. JUNO will determine the neutrino mass ordering [24] in 2026–27; DUNE and Hyper-Kamiokande will resolve the θ_{23} octant around 2030; nEXO will search for $0\nu\beta\beta$ around 2030; Belle II will measure the τ mass to the precision needed to resolve the Koide deviation. If inverted ordering is observed, if the first octant ($\sin^2 \theta_{23} < 0.5$) is confirmed, or if $0\nu\beta\beta$ is detected, the theory is falsified.

Four assumptions used in this paper—the tensor product, $\mu^2 < 0$, v_{EW} , and confinement—are not yet derived from A1–A3. Paper II [19] derives the first two and proves why $V = -H$ is the unique potential. Paper III [20] develops the cosmological consequences.

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